

下面這版我會當成 **HiABC Proposal** 精要版，適合先給團隊討論、做院內 proposal synopsis，之後再展開成 IRB / grant protocol。

# HiABC Trial

## Hypertonic Saline and Active Cycle of Breathing Technique in Bronchiectasis Care: A Multicenter Randomized Controlled Trial

### 1. Background and rationale

Bronchiectasis is characterized by impaired mucociliary clearance, sputum retention, chronic airway infection and recurrent pulmonary exacerbations. Airway-clearance therapy is therefore a cornerstone of management, but evidence supporting specific long-term strategies remains limited.

The 2026 CHEST bronchiectasis guideline conditionally recommends both **nebulized hypertonic saline** and **breathing techniques including ACBT**, but the certainty of evidence remains very low for hypertonic saline and low for ACBT. Chest Journal

In the 2025 CLEAR trial, 288 patients with bronchiectasis were followed for 52 weeks. Hypertonic saline was associated with a numerically lower exacerbation rate than no hypertonic saline, **0.76 vs 0.98 exacerbations/person-year**, although the difference did not reach statistical significance (adjusted difference -0.25; 95% CI -0.57 to 0.07; P=0.12). 新英格蘭醫學雜誌 +1

Meanwhile, a recent randomized Respiriology study comparing airway-clearance techniques found that ACBT improved selected HRQoL measures and showed an exploratory signal toward a longer time to subsequent exacerbation: mean time to first exacerbation was **158 days with ACBT**, compared with 118 days with walking+huffing and 89 days with OPEP, although this finding was limited by loss to follow-up and was not definitive. Wiley Online Library

These findings support a complementary therapeutic concept:

**3% hypertonic saline → airway hydration and mucus mobilization → ACBT → active secretion clearance**

Whether this **combined airway-clearance strategy** can delay bronchiectasis exacerbations and improve symptoms has not been adequately tested.

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### 2. Study hypothesis

In adults with productive bronchiectasis at increased risk of exacerbation, a structured airway-clearance program consisting of **3% hypertonic saline followed by ACBT** will prolong the time to first pulmonary exacerbation compared with **0.9% Isotonic Saline + Pragmatic Usual BE Care**.

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### 3. Study design

**Prospective, multicenter, randomized, controlled, parallel-group trial with blinded endpoint assessment.**

**Randomization: 1:1**

| <b>Control: 0.9% Isotonic Saline + Pragmatic Usual BE Care</b> | <b>HiABC intervention</b>                               |
|----------------------------------------------------------------|---------------------------------------------------------|
| <b>0.9% isotonic saline 5 mL QD</b>                            | <b>3% hypertonic saline 5 mL QD → standardized ACBT</b> |
| 6 days/week                                                    | 6 days/week                                             |
| Pragmatic usual BE care                                        | Usual BE care                                           |

### **Control arm description**

**Control = 0.9% Isotonic Saline + Pragmatic Usual BE Care.**

- Nebulized 0.9% isotonic saline 5 mL once daily, 6 days/week, for 52 weeks, using the same nebulizer type and administration schedule as the intervention arm.
- **Pragmatic usual BE care:** all other bronchiectasis management is left to the treating physician according to routine clinical practice and current guidelines (for example, maintenance macrolide, inhaled antibiotic, bronchodilator and treatment of exacerbations). Participants may continue their pre-existing, self-directed airway-clearance habits.
- The control arm does **not** receive structured ACBT training, technique review or supervised ACBT sessions.
- Background treatments and any changes during follow-up are prospectively recorded in both arms.

Treatment duration and follow-up: **52 weeks.**

The trial evaluates the **combined HiABC strategy**, not the independent contribution of HS versus ACBT.

## **4. Study population**

### **Major inclusion criteria**

- Age  $\geq 18$  years
- CT-confirmed bronchiectasis
- Chronic productive cough / sputum production
- Clinically stable  $\geq 4$  weeks
- **$\geq 2$  physician-treated pulmonary exacerbations during the preceding 12 months**
- Able to perform standardized ACBT

- **No initiation or major change of multidrug anti-NTM therapy within the preceding 8 weeks**  
(近 8 週內未剛開始或大幅調整多藥抗 NTM 治療)

Using  $\geq 2$  prior exacerbations enriches the population for first-AE events and makes a time-to-event primary endpoint more feasible.

### Major exclusion criteria

- Cystic fibrosis
- Current acute exacerbation
- Current regular hypertonic saline use
- Current regular structured ACBT or equivalent intensive ACT
- Major/recent hemoptysis
- Inability to tolerate 3% HS
- Unstable asthma / significant bronchospasm

### NTM-related exclusions (strictly applied / 嚴格排除):

- Multidrug NTM treatment **newly started** within 4–8 weeks before enrollment  
(收案前 4~8 週內才「剛開始」啟動 NTM 多藥治療者)
- A major change in NTM drug treatment, or surgical resection, **expected** during the 52-week study follow-up  
(預期在試驗隨訪期間 (52 週) 即將進行重大 NTM 藥物調整或外科手術切除者)
- Active cavitory disease with a **recent history of major hemoptysis**  
(合併活動性空洞型且近期有大咳血病史者)

Stable long-term macrolide, inhaled antibiotic, bronchodilator or other guideline-based therapy is allowed.

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## 5. Primary endpoint

### Time to first pulmonary exacerbation through Week 52

Pulmonary exacerbation should use a prespecified consensus definition requiring deterioration in respiratory symptoms for  $\geq 48$  hours together with clinician-directed treatment, preferably systemic antibiotics.

Events should be reviewed by a blinded exacerbation adjudication committee.

Time-to-first exacerbation has precedent as a primary endpoint in bronchiectasis RCTs, including the WILLOW brensocatib study. 新英格蘭醫學雜誌

Primary analysis:

**Kaplan–Meier analysis + Cox proportional-hazards model**

Report HR, 95% CI and two-sided P value.

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## 6. Key secondary outcomes

- Annualized pulmonary exacerbation rate over 52 weeks
- Proportion experiencing  $\geq 1$  exacerbation
- Time to first severe exacerbation / hospitalization
- Total antibiotic-treated exacerbation days
- QoL-B Respiratory Symptoms score
- Bronchiectasis Health Questionnaire
- Leicester Cough Questionnaire
- Sputum burden / ease of expectoration
- Lung function
- Sputum bacterial and AFB microbiology
- Treatment adherence
- Adverse events and treatment intolerance

QoL assessments can be performed at baseline, Month 3, Month 6 and Month 12.

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## 7. ACBT intervention

3% HS inhalation is followed immediately by approximately **10–20 minutes** of standardized ACBT:

**breathing control → thoracic expansion → breathing control → forced expiration/huff → repeat as required.**

Training should be standardized across centers using trained respiratory therapists/physiotherapists and preferably a common instructional video.

Technique review:

baseline → Week 2 → Month 1 → Month 3 → as required.

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## 8. Background treatment and macrolide use

Maintenance macrolides should not be excluded, because frequent exacerbators commonly receive them in contemporary practice.

Preferably:

**chronic macrolide or inhaled antibiotic therapy should be stable for  $\geq 8$ –12 weeks before randomization.**

Randomization should balance:

- study center
- maintenance macrolide use: yes/no
- prior exacerbation burden

- chronic *Pseudomonas aeruginosa*: yes/no

Any clinically necessary treatment escalation during follow-up is permitted and prospectively recorded. Primary analysis remains intention-to-treat.

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## 9. Sample size

Because the primary endpoint is time to first exacerbation, the required sample depends strongly on the expected treatment HR and the proportion developing an event during 52 weeks.

A reasonable initial planning assumption for this enriched population is:

- two-sided  $\alpha = 0.05$
- power = 80%
- expected HR  $\approx 0.60$ – $0.65$
- control 12-month first-exacerbation risk  $\approx 65$ – $70\%$
- attrition  $\approx 10\%$

Under these assumptions, a practical target is approximately:

**N = 240–300**

For protocol development, I would initially propose:

**N = 240 participants, 120 per arm**

but explicitly state that the final sample size will be refined using local bronchiectasis cohort data on 12-month first-exacerbation probability.

If a more conservative HR of  $\sim 0.65$  is chosen, the final N may need to approach 280–300.

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## 10. Follow-up schedule

Screening → Randomization → Week 2 → Month 1 → Month 3 → Month 6 → Month 9 → Month 12

Monthly remote contact can capture:

- exacerbations
  - antibiotics
  - hospitalization
  - adherence
  - treatment changes
  - adverse events
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## 11. Proposed primary objective

To determine whether a structured airway-clearance strategy combining nebulized 3% hypertonic saline with ACBT prolongs the time to first pulmonary exacerbation compared with nebulized 0.9% Isotonic Saline + Pragmatic Usual BE Care in adults with productive bronchiectasis and recurrent exacerbations.

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## 12. Proposed key message

HiABC tests an integrated airway-clearance strategy rather than an individual component. CLEAR demonstrated a numerical but inconclusive exacerbation signal with hypertonic saline, while emerging randomized evidence supports ACBT as an airway-clearance technique. HiABC will determine whether combining mucus hydration with active secretion clearance translates into clinically meaningful prevention of bronchiectasis exacerbations.

我覺得這個版本最值得往下發展的是 **N≈240、52 weeks、time to first AE primary、annualized AE rate key secondary**；研究問題清楚，也比把 QoL 當 primary 更有 clinical impact。